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SIT-DOWN COUNTERBALANCED FORKLIFT TRUCK WITH A LOW CENTER OF GRAVITY

Abstract

A sit-down counterbalanced forklift truck comprises a chassis, chassis compartment, power plant, plurality of wheels, lift assembly, operator compartment, and counterweight. The chassis has front, rear, left, and right sides. The chassis compartment is situated between those sides. The power plant is located in the chassis compartment. At least one of the plurality of wheels is operably connected to the power plant to drive the truck. The lift assembly is attached to the chassis, has forks configured to lift a load, and is operably connected to the power plant. The operator compartment is above and attached to the chassis, and has an operator seat and an operator compartment floor. The truck has a center of gravity situated at a gravitational center height that is lower than or equal to a height of at least one of the plurality of wheels.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation and claims the benefit, under 35 U.S.C. § 120, of U.S. patent application Ser. No. 19/171,420 entitled “Sit-Down Counterbalanced Forklift Truck Having an Operator Compartment with a Broad Flat Floor,” filed Apr. 7, 2025, which in turn claims the benefit, under 35 U.S.C. § 120, of U.S. patent application Ser. No. 18/640,146, entitled “Sit-Down Counterbalanced Forklift Truck Based on a Common Chassis,” filed Apr. 19, 2024 now U.S. Pat. No. _____, which in turn claims the benefit, under 35 U.S.C. § 120, of U.S. patent application Ser. No. 17/845,030, entitled “Lift Truck Having Advantageous Design Elements,” filed Jun. 21, 2022, now U.S. Pat. No. 12,012,321, which in turn claims the benefit, under 35 U.S.C. § 120, of U.S. patent application Ser. No. 16/424,998, entitled “Lift Truck Having Advantageous Design Elements,” filed May 29, 2019, now U.S. Pat. No. 11,427,450, which in turn claims priority under 35 U.S.C. § 119(e) to U.S. Provisional Patent Application No. 62/679,686, entitled “Modular Lift Truck Design Based on a Common Chassis,” filed Jun. 1, 2018. The entire disclosures of those foregoing patent applications are incorporated by reference herein.

TECHNICAL FIELD

[0002] The field of this disclosure relates generally to materials-handling vehicles. More specifically, this disclosure relates to lift trucks having different operator-accessible and operator-inaccessible compartments.

BACKGROUND INFORMATION

[0003] Lift trucks, including forklift trucks, are commonly used in manufacturing and warehousing facilities to lift and to transport materials. Lift trucks incorporate multiple components such as counterweights, a hydraulic system, electronics, an energy source, wheels (and tires), a steering mechanism, an operator's seat within an operator compartment, and a lift assembly, all supported by the truck's chassis. These lift trucks are ideally small and agile, have a small turning radius and compact footprint, and are able to repeatedly lift thousands of pounds of materials with little maintenance. An operator (also referred to as a “driver”) controls the lift truck from the operator compartment, often for multiple hours at a time, and may frequently enter and exit the operator compartment of the truck to manually handle or adjust the materials that are being transported. The operator's comfort within the operator compartment, visibility from the operator compartment, and ease of physically accessing the operator compartment are relevant to efficient material transport.

[0004] To improve maneuverability, the design of lift trucks often will compromise the ergonomics of the operator compartment. For example, the truck components are commonly arranged in a stacked vertical orientation to minimize the truck's length and width. This arrangement can result in a lift truck having a high operator compartment floor, thereby requiring an operator to use multiple

vertical steps to enter and to exit the truck. The high floor height of the operator compartment relative to the ground is one shortcoming of commonly available lift trucks and makes it unduly burdensome for an operator to quickly and easily step into and out of the truck. A high floor height also raises the elevation of truck components that are located on or above the floor, such as overhead guards, operator seats, operator compartments, or portions of the lift assembly. Such elevated placement can hamper visibility if the view from the operator's seat is limited; hinder the operator when moving into, out of, or within the operator compartment; and undesirably raise the center of gravity of the lift truck. Another shortcoming with known lift trucks is the limited floor space within the operator compartment. Columns and foot controls that extend from the floor can restrict movement of the operator's legs and feet while using the truck. In the absence of ergonomically designed lift trucks, operator comfort and productivity may suffer.

OVERVIEW OF DISCLOSURE

[0005] One aspect of this disclosure relates to lift trucks having a modular design that incorporates a common chassis, which can accommodate one of a variety of hydraulic systems, electronics, energy sources, counterweights, wheels, steering, seating and floor components assembled in various configurations, to provide a driver with ergonomic and productive truck operation.

[0006] One aspect of an ergonomic operator compartment can be achieved by placing the floor of the operator compartment relatively close to the ground, such that the operator need only take a single vertical step to enter and to exit the compartment comfortably. A flat and broad floor in the operator compartment can also improve operator comfort. In addition, it is beneficial for the floor to have few or no obstacles that the operator has to avoid or to maneuver around when getting into and out of the compartment, or while driving the lift truck. For example, the operator compartment may be free of a steering wheel and/or foot controls, instead having means for steering and/or control integrated into the operator's seat.

[0007] It is also often desirable to design a lift truck so that the lift truck driver has views from the operator's seat with minimal or no obstructions. This can be achieved, for example, by designing truck components to have a low profile or to slope downwardly at a specific location, such as behind the operator's seat to allow for improved rear visibility by the operator. The operator's seat itself can be positioned within the operator compartment in a manner that provides sufficient leg room for rapid ingress and egress, as well as during routine truck driving. In addition, the operator's seat can move in a manner that provides the operator with unobstructed views both in front of and behind the lift truck without incurring undue eye, neck, back, or leg strain. For example, the operator's seat can rotate about a vertical axis and be designed to slide forward and backward independent from, or concurrent with, such rotation. The seat may incorporate a mechanism that adjusts the seat height vertically, in addition to or instead of the rotating and/or sliding mechanisms.

[0008] The chassis of the lift truck can be designed to include compartments that contain specific components, which can be placed in various locations within the truck. For example, a chassis compartment can contain an energy source, such as one or more batteries, and/or an electric motor, or hydraulic components that are used for the lift assembly. A chassis compartment can be designed to be readily accessible to an operator, such as to allow for routine maintenance of a component, or it can be designed to limit accessibility by an operator. An operator-inaccessible compartment can contain certain components of the lift truck that generally require little maintenance, such as service-free batteries, motors, and/or hydraulic components.

[0009] A lift truck generally includes a number of standard components. For example, lift trucks have an energy source that provides energy to the truck, such as one or more lithium-ion or lead-acid batteries. The lift truck incorporates a lift assembly with an attachment, such as forks, used to vertically lift a load. Lift trucks also have a plurality of wheels that are used to move the truck forward and backward. For example, a lift truck may have three wheels, or it may have four wheels, or it may have more than four wheels. The lift truck can include a roof over the operator compartment, such as an overhead guard. Truck components can be arranged in a horizontal

manner, rather than vertically, using a prudently designed chassis. One will appreciate, however, that vertical arrangements of truck components using other advantageous configurations described herein are also possible, as are mixed horizontal and vertical arrangements.

[0010] In one embodiment, a battery-powered sit-down counterbalanced forklift truck for use in an indoor warehouse on a warehouse floor comprises a chassis, a chassis compartment, a battery, a motor, a plurality of wheels, a counterweight, an operator compartment, an operator compartment floor, and an operator seat. The chassis has a front side, a rear side, a left side, and a right side. The chassis compartment is situated between the front side and the left side of the chassis and between the left side and right side of the chassis. The battery is located in the chassis compartment. The motor is located in the chassis compartment and operably connected to the battery. At least one of the plurality of wheels is operably connected to the motor to drive the truck. The plurality of wheels are sized and placed to provide the truck with a footprint and maneuverability suitable for use in a warehouse environment. The counterweight is attached to the rear side of the chassis. The operator compartment is above the chassis and attached to the chassis. The operator compartment floor is above the chassis compartment and below the operator compartment. The operator compartment floor is substantially flat, situated between the left side of the chassis and the right side of the chassis, and extends substantially all the way between the left side of the chassis and the right side of the chassis. The operator seat is in the operator compartment and comprises a backrest. The operator compartment floor extends rearward at least partially under the backrest of the operator seat.

[0011] In some additional, alternative, or selectively cumulative embodiments, the truck further comprises a drive axle attached to one or more of the plurality of wheels, wherein the motor is located in the drive axle.

[0012] In some additional, alternative, or selectively cumulative embodiments, the truck further comprises a hydraulic system in the chassis compartment. The hydraulic system is operably connected to the battery and comprises a hydraulic pump, a motor configured to operate the hydraulic pump, a hydraulic tank fluidly connected to the hydraulic pump, and a hydraulic valve. The hydraulic pump is directly connected to the hydraulic tank without a hose.

[0013] In some additional, alternative, or selectively cumulative embodiments, the chassis compartment is an operator-inaccessible compartment.

[0014] In some additional, alternative, or selectively cumulative embodiments, the operator compartment floor is between approximately 24 centimeters and approximately 76 centimeters above the ground.

[0015] In some additional, alternative, or selectively cumulative embodiments, the operator compartment floor is between approximately 43 centimeters and approximately 56 centimeters above the ground.

[0016] In some additional, alternative, or selectively cumulative embodiments, one or more of the plurality of wheels has a wheel height, and the operator compartment floor has a floor height that is less than or equal to 90% the wheel height.

[0017] In some additional, alternative, or selectively cumulative embodiments, one or more of the plurality of wheels has a wheel height and a wheel radius height, and the operator compartment floor has a floor height that is between the wheel height and the wheel radius height.

[0018] In some additional, alternative, or selectively cumulative embodiments, the truck further comprises a lift assembly attached to the chassis and having forks.

[0019] In some additional, alternative, or selectively cumulative embodiments, the chassis compartment has a chassis compartment top area between front wheels and the counterweight and between the left side and right side of the chassis. The operator compartment floor has a floor area that is greater than or equal to 90% of the chassis compartment top area.

[0020] In some additional, alternative, or selectively cumulative embodiments, the chassis compartment has a chassis compartment top area between front wheels and the counterweight and

between the left side and right side of the chassis. The operator compartment floor has a floor area that is greater than or equal to 75% of the chassis compartment top area.

[0021] In some additional, alternative, or selectively cumulative embodiments, the operator compartment floor has a floor area that is substantially flat for greater than or equal to 50% of an area of a top of the chassis compartment.

[0022] In some additional, alternative, or selectively cumulative embodiments, the truck further comprises an operator-accessible compartment having a top cover, and an operator-serviceable component located in the operator-accessible compartment. The operator-accessible compartment is located above the counterweight.

[0023] In some additional, alternative, or selectively cumulative embodiments, the operator-serviceable component is one of a motor controller, fuse, VSM, contactor, or any combination thereof.

[0024] In some additional, alternative, or selectively cumulative embodiments, the counterweight comprises multiple distinct pieces.

[0025] In some additional, alternative, or selectively cumulative embodiments, the battery is a service-free battery, such as, for example, a lithium-based battery.

[0026] In some additional, alternative, or selectively cumulative embodiments, there is no intermediate step on the truck between the warehouse floor and the operator compartment floor to facilitate entry into or exit from the operator compartment by an operator.

[0027] In some additional, alternative, or selectively cumulative embodiments, the entire backrest of the operator seat is above the operator compartment floor.

[0028] In some additional, alternative, or selectively cumulative embodiments, the operator seat is rotatable about a vertical axis.

[0029] In some additional, alternative, or selectively cumulative embodiments, the operator seat has a range of rotation of greater than or equal to approximately ± 180 degrees from a straight-forward-facing direction.

[0030] In some additional, alternative, or selectively cumulative embodiments, the operator seat has a range of rotation of greater than or equal to approximately ± 22 degrees from a straight-forward-facing direction.

[0031] In some additional, alternative, or selectively cumulative embodiments, the operator seat is configured to slide backward as the operator seat rotates, so as to keep the operator's legs within an operator compartment as the operator seat rotates.

[0032] In some additional, alternative, or selectively cumulative embodiments, the operator seat has a bottom having front, left, right, and rear edges, and the operator compartment floor extends substantially flat front-to-back from a front cowl proximate the front of the truck to a rear floor edge behind a front edge of the seat bottom.

[0033] In some additional, alternative, or selectively cumulative embodiments, the truck has no steering wheel extending from the operator compartment floor.

[0034] In some additional, alternative, or selectively cumulative embodiments, the truck has no foot controls.

[0035] In some additional, alternative, or selectively cumulative embodiments, the operator seat comprises at least one armrest, and wherein operator controls are located on said at least one armrest.

[0036] In some additional, alternative, or selectively cumulative embodiments, the plurality of wheels comprises at least one left wheel and at least one right wheel, and the operator compartment floor extends side-to-side from the at least one left wheel to the at least one right wheel.

[0037] the truck has a center of gravity situated at a gravitational center height that is lower than or equal to a drive wheel height of the plurality of wheels.

[0038] the truck has a center of gravity situated at a gravitational center height that is lower than or equal to a floor height of the operator compartment.

[0039] In some additional, alternative, or selectively cumulative embodiments, the operator compartment is equally accessible to an operator from both the right and left sides of the chassis.

[0040] In some additional, alternative, or selectively cumulative embodiments, the operator seat comprises a bottom pad, and the operator compartment floor such that the entire bottom pad of the operator seat is above the operator compartment floor.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0041] FIG. 1 is an isometric drawing of a lift truck from a front right perspective, according one embodiment.

[0042] FIG. 2 is an isometric drawing of the lift truck of FIG. 1 from a rear left perspective.

[0043] FIG. 3 is a right side view of the lift truck of FIG. 1.

[0044] FIG. 4 is a left side view of the lift truck of FIG. 1.

[0045] FIG. 5 is a top view of the lift truck of FIG. 1.

[0046] FIG. 6 is an isometric drawing of a lift truck having seatside steering from a front right perspective, according to one embodiment.

[0047] FIGS. 7A and 7B are isometric drawings of the lift truck of FIG. 6 from a rear left perspective, illustrating an exemplary rear compartment in closed (FIG. 7A) and open (FIG. 7B) configurations.

[0048] FIG. 8 is a left side view of the lift truck of FIG. 6.

[0049] FIG. 9 is a right side view of the lift truck of FIG. 6.

[0050] FIGS. 10A, 10B, 10C and 10D illustrate embodiments of seatside steering controls that can be used for lift trucks, including the truck of FIG. 6. FIG. 10A is an isometric drawing of a lift truck from a rear right perspective, according to one embodiment. FIG. 10B is an isometric drawing of the lift truck of FIG. 10A from a front right perspective. FIG. 10C is a right side view of the lift truck of FIG. 10A. FIG. 10D is an isometric drawing of a lift truck from a front left side perspective, enlarged to show greater details according to one embodiment.

[0051] FIG. 11 is a front view of the lift truck of FIG. 6.

[0052] FIG. 12 is a top view of the lift truck of FIG. 6.

[0053] FIG. 13 is an isomeric drawing of the chassis of a lift truck from a left front perspective, according to one embodiment.

[0054] FIG. 14 is an illustration of an embodiment of a lift truck having a lithium-ion-battery-based energy source and seatside steering, in a partly exploded left side view, with an operator compartment (top) shown elevated above the battery bank (center) and chassis (bottom).

[0055] FIG. 15A is a top view of the battery bank and chassis of an embodiment of the lift truck of FIG. 13.

[0056] FIG. 15B is top view of the battery bank and chassis of an alternative embodiment of the lift truck of FIG. 13.

[0057] FIG. 16 is an isometric drawing of a lift truck having a lead-acid-battery-based energy source and seatside steering, according to one embodiment, from a left front perspective.

[0058] FIG. 17 is a partly exploded left side view of the lift truck of FIG. 16, with an operator compartment (top) shown elevated above a lead-acid battery (center) and chassis (bottom).

[0059] FIG. 18A is a top view of the battery and chassis of the lift truck of FIG. 17.

[0060] FIG. 18B is a top view of the battery and chassis of an alternative embodiment of the lift truck of FIG. 17, with the fork carriage and mast removed.

[0061] FIG. 19 is an isometric drawing of a lift truck with a hydrogen-fuel-cell-based energy source and seatside steering, according to one embodiment, from a left front perspective.

[0062] FIG. 20 is a partly exploded left side view of the lift truck of FIG. 19, with an operator

compartment (top) shown elevated above a hydrogen-fuel-cell-based energy source (center) and chassis (bottom).

[0063] FIG. **21A** is a top view of the energy source and chassis of the lift truck of FIG. **20**.

[0064] FIG. **21B** is a top view of the energy source and chassis of an alternative embodiment of the lift truck of FIG. **20**, with the fork carriage and mast removed.

[0065] FIG. **22** is an isometric drawing of a lift truck with an internal-combustion-engine-based energy source and seatside steering, according to one embodiment, from a left front perspective.

[0066] FIG. **23** is a partly exploded left side and top view of the lift truck of FIG. **22**, with an operator compartment (top) shown elevated above an internal-combustion-engine-based energy source (center) and chassis (bottom).

[0067] FIG. **24A** is a top view of the energy source and chassis of the lift truck of FIG. **23**.

[0068] FIG. **24B** is an additional illustration of FIG. **24A**, with the fork carriage and mast removed.

[0069] FIGS. **25A** and **25B** (collectively FIG. **25**) are respective rear left and front left partly cutaway views that illustrate exemplary counterweight designs that can be used for a lift truck, which are able to accommodate different energy source and drive components and arrangements, according to various embodiments.

[0070] FIG. **26** is an isometric drawing of a lift truck having a different form factor than the lift truck of FIG. **1**, from a rear left perspective, according to one embodiment.

[0071] FIG. **27** is a right side view of a lift truck having an additional different form factor than the lift truck of FIG. **1**, according to one embodiment.

[0072] FIG. **28** is an isometric drawing of a lift truck with a short cowl and seatside steering, according to one embodiment, from a left rear perspective.

[0073] FIG. **29** is an isometric drawing of a lift truck with the mast and fork chassis removed, according to one embodiment, from a right front perspective.

[0074] FIG. **30** is an isometric drawing of an additional lift truck with the mast and fork chassis removed, according to one embodiment, from a right front perspective.

[0075] FIG. **31** is a right side view of a lift truck, comparing a sight line of an old lift truck to a sight line achievable according to one embodiment of a lift truck disclosed herein.

[0076] FIG. **32** is a top view of a lift truck, comparing floor area of an old lift truck to floor area according to one embodiment of a lift truck disclosed herein.

DETAILED DESCRIPTION OF EMBODIMENTS

[0077] Example embodiments are described below with reference to the accompanying drawings. Unless otherwise expressly stated, the sizes, positions, etc., of components, features, elements, etc., as well as any distances therebetween, are not necessarily to scale, and may be disproportionate and/or exaggerated for clarity.

[0078] The terminology used herein is for the purpose of describing particular example embodiments only and is not intended to be limiting. As used herein, the singular forms “a,” “an” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It should be recognized that the terms “comprise,” “comprises,” “comprising,” “include,” “includes,” “including,” “has,” “have,” and “having,” when used in this document, specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof. Unless otherwise specified, a range of values, when recited, includes both the upper and lower limits of the range, as well as any sub-ranges therebetween. Unless indicated otherwise, terms such as “first,” “second,” etc., are only used to distinguish one element from another and not to imply any relative order, placement, or ranking. For example, one element could be termed a “first element” and similarly, another element could be termed a “second element,” or vice versa. The section headings used herein are for organizational purposes only and are not to be construed as limiting the subject matter described.

[0079] Unless indicated otherwise, the terms “about,” “thereabout,” “substantially,” etc. mean that

amounts, sizes, formulations, parameters, and other quantities and characteristics are not and need not be exact, but may be approximate and/or larger or smaller, as desired, reflecting tolerances, conversion factors, rounding off, measurement error and the like, and other factors known to those of skill in the art.

[0080] Spatially relative terms, such as “right,” “left,” “below,” “beneath,” “lower,” “above,” and “upper,” and the like, may be used herein for ease of description to describe one element's or feature's relationship to another element or feature, as illustrated in the drawings. It should be recognized that the spatially relative terms are intended to encompass different orientations in addition to the orientation depicted in the drawings. For example, if an object in the figures is turned over, elements described as “below” or “beneath” other elements or features would then be oriented “above” the other elements or features. Thus, the term “below” can, for example, encompass both an orientation of above and below. An object may be otherwise oriented (e.g., rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may be interpreted accordingly.

[0081] Unless clearly indicated otherwise, all connections and all operative connections may be direct or indirect. Similarly, unless clearly indicated otherwise, all connections and all operative connections may be rigid or non-rigid.

[0082] Like numbers refer to like elements throughout. Thus, the same or similar numbers may be described with reference to other drawings even if they are neither mentioned nor described in the corresponding drawing. Also, even elements that are not denoted by reference numbers may be described with reference to other drawings. Additionally, the drawings may include non-essential elements that are included only for the sake of thoroughness. These non-essential elements may be removed entirely or left only in outline form if drawing changes are desired to create greater clarity.

[0083] The embodiments described herein are merely examples, set forth by way of illustration only and not limitation. Those skilled in the art will recognize in light of the teachings herein that there are alternatives, variations and equivalents to the example embodiments described herein and their component parts. For example, other embodiments are readily possible, variations can be made to the embodiments described herein, and there may be equivalents to the components, parts, or steps that make up the described embodiments.

[0084] For the sake of clarity and conciseness, certain aspects of components or steps of certain embodiments are presented without undue detail where such detail would be apparent to those skilled in the art in light of the teachings herein and/or where such detail would obfuscate an understanding of more pertinent aspects of the embodiments.

[0085] As one skilled in the art will appreciate in light of this disclosure, certain embodiments of the lift trucks and designs therefor that are disclosed herein may be capable of achieving certain advantages, including, in some cases, one or more of the following: (1) an operator compartment with a low floor to facilitate comfortable entry into the operator compartment, exit from the operator compartment, and operation of the truck within the operator compartment; (2) an operator compartment having few or no obstacles to operator movement, especially along the floor of the operator compartment; (3) an operator-inaccessible chassis compartment that inhibits operators from disturbing high-reliability components that should ideally be accessed only by specifically trained service personnel; (4) a low-profile design to facilitate operator visibility; (5) a low center of gravity for enhanced stability; (6) a multi-piece counterweight of different densities; (7) a modular chassis design having form factors that are adaptable to multiple operator uses and to accommodate one of multiple possible energy sources, power plants, and/or other components, thereby providing manufacturers of lift trucks design options to allow for placement of various combinations of these components in various locations, while also being tailored to a customer's specific use; and (8) an operator compartment with a configurable storage system for tools and devices needed to perform daily tasks. The foregoing and other advantages of various embodiments will be apparent upon reading this document.

[0086] FIGS. **1** and **2** are front and rear isometric illustrations, respectively, of a lift truck **100** according to one example embodiment. Referring to FIG. **1**, the lift truck **100** can include an operator compartment **105** and a mast **110** having mast columns **115**. Any type of mast may be used in the lift trucks disclosed herein. For example, the mast **110** may be an embodiment of the masts disclosed in U.S. Patent Appl. Publ. No. US 2017/0073203, the entire disclosure of which is incorporated by reference herein.

[0087] The lift truck **100** has a chassis **130** below the operator compartment **105** having an operator's seat **140** and an operator compartment floor **150** therein. The chassis **130** can be designed to be modular so as to accommodate a variety of components in assorted locations in order to meet an operator's specific needs. Thus, the chassis **130** can include compartments that are designed to contain certain components and that can be placed in certain locations within the lift truck **100**. Additionally, the operator compartment floor **150** can be low and spacious and provide for a configurable area **152** within the operator compartment **105**. The configurable area **152** may include a shelf or other means for storage on the inside of the operator compartment **305**, such as a cup holder, a holder for a computerized device (such as a GPS device or an inventory device), miniature refrigerator or other food storage compartment, or other compartment(s) suitable for storing other operator items, such as tools or safety vests or jackets.

[0088] The chassis **130** can have two sides and a bottom that are configured to define a chassis compartment that is within the two sides and above the bottom. The chassis compartment can extend between the front end of the chassis, such as between the drive axle **324** (FIG. **10B**) (which may extend along an axis (or a centerline) between the wheel centers of the drive wheels), to a counterweight **112** that is attached to the rear side, or at the rear end, of the chassis **130**. The location of the front and rear ends of the chassis **130** are in reference to an operator seated in the seat **140** in the operator compartment **105**, looking straight forward toward the mast **110** along a longitudinal axis **125**. The chassis compartment can extend between the left side and the right side of the chassis, that is, completely across the width of the lift truck **100**, or it may extend between a subset of the width of the lift truck **100**. The chassis compartment can contain an energy source, such as, for example, one or more batteries, and/or an electric motor, as well as hydraulic components that are used for the lift assembly, such as a hydraulic motor, hydraulic tank and/or at least one hydraulic valve.

[0089] The chassis **130** can include one or more counterweights **112** (shown in FIG. **2**) made of one or more pieces, with each piece having one or more properties such as tensile strength and density. A counterweight **112** that is formed from, or made with, multiple counterweight pieces may in some circumstances be considered to be one functional unit that may be referred to herein as simply a "counterweight." Thus, the term "counterweight" may mean a single counterweight or a functional unit comprising multiple counterweight pieces, depending on context. Optionally, a chassis compartment may contain one or more counterweights **112**. For example, the counterweight **112** may be a two-piece counterweight comprising multiple SAE J431 Gray Iron Casting grades. The counterweight **112** may be attached to the side of the chassis compartment distal from the mast, that is, it may be attached to the rear side or rear end of the chassis.

[0090] A chassis compartment can be located, for example, under the operator compartment floor **150** and/or behind the seat **140**. The members of the chassis **130** may define a single chassis compartment or multiple chassis compartments depending on the design of the chassis **130**. For example, if the primary structural members of the chassis are on the sides and bottom, then a single chassis compartment may be defined. Alternatively, if there are lateral and/or longitudinal reinforcements across the chassis, then there may be multiple chassis compartments.

[0091] A plate may be used to form a cover, such as a top cover, of a chassis compartment. The plate can be designed not to be readily removable by an operator, to form a chassis compartment that is not readily accessible to the operator when the plate is on the chassis, thereby forming an operator-inaccessible chassis compartment. The operator compartment floor **150** can be wholly or

partially made from a plate. The location of an inaccessible compartment can be, for example, under the floor **150** of the operator compartment **105**. An inaccessible compartment may be useful for containing truck components that do not often require maintenance, such as a service-free battery, which may be under a manufacturer's warranty or require specialized training to service. In an embodiment, an inaccessible chassis compartment contains an energy source including a service-free battery. The energy source can provide power to turn the wheels of the truck, to actuate the lift assembly, or to both turn the wheels and actuate the lift assembly.

[0092] The lift truck **100** incorporates an energy source that stores energy for the truck. The energy from the energy source may be converted to power by a power plant to turn one or more of a plurality of wheels to propel the truck forward and backward and to actuate the lift assembly. The energy source used for turning the wheels and propelling the truck may be the same as, or may be different from, the energy source used for the lift assembly. The term “energy source” may be used herein to convey one or more of a fuel, a fuel conversion device, a battery, and a battery bank. The power plant may include, but is not limited to, an electric motor, an internal combustion engine, a hydrogen fuel cell, or any combination of the foregoing. For example, the energy source used to propel and steer the truck may be a lithium-ion battery or a lead-acid battery or a bank of multiple such batteries, and the lift assembly and/or wheels may be actuated directly or indirectly by an electric motor powered by the one or more batteries. In another embodiment, the energy source is a hydrogen fuel cell that runs an electric motor and/or an electric pump. In certain embodiments, the energy source includes a lithium-ion battery or a hydrogen fuel cell. In some contexts, “fuel” may refer to electricity, fossil fuel, such as gasoline or diesel, or battery compositions, such as lead-acid or lithium-ion, or hydrogen, such as liquid or gaseous. In some contexts, “power plant” may refer to a fuel cell or a motor or an internal combustion engine or any device that converts energy into physical motion of the lift truck or its hydraulic system.

[0093] The energy source can be partially or wholly located within one or more chassis compartments, such as having a battery bank, electric motor, and a hydraulic system for the lift assembly in an operator-inaccessible chassis compartment and electronic components (for example, power electronics such as motor controllers, fuses, a vehicle controller such as a vehicle system module or “VSM”, and contactors) in an operator-accessible compartment. All or some of the components of the energy source may be service-free, or at least one of the components may require service on a regular basis, such as annually or biannually. The energy source may be a low-service or service-free battery or battery bank, such as a lithium-ion battery or battery bank. A battery bank of multiple batteries may in some circumstances be considered to be one functional unit that may be referred to herein as simply a “battery.” Thus, the term “battery” may mean a single battery or a functional unit comprising multiple batteries, depending on context.

[0094] The operator's seat **140** can be rotatable about a vertical axis **145** to provide an operator with views behind and in front of the truck by rotating the seat. For example, the operator seat **140** may have a range of rotation of greater than or equal to approximately ± 22 degrees from a straight-forward-facing direction. The operator seat **140** may have a range of rotation of greater than or equal to approximately ± 30 degrees from a straight-forward-facing direction. The operator seat **140** may have a range of rotation of greater than or equal to approximately ± 60 degrees from a straight-forward-facing direction. The operator seat **140** may have a range of rotation of greater than or equal to approximately ± 90 degrees from a straight-forward-facing direction. The operator seat **140** has a range of rotation of greater than or equal to approximately ± 180 degrees from a straight-forward-facing direction. The operator seat **140** may have a range of rotation of greater than or equal to approximately ± 270 degrees from a straight-forward-facing direction. The operator seat **140** may have a range of rotation of approximately ± 360 degrees from a straight-forward-facing direction.

[0095] Depending on the layout of other components in the operator compartment **105** or cost considerations, the operator seat **140** may intentionally be limited to a range of rotation that is less

than or equal to approximately ± 270 degrees, ± 180 degrees, ± 90 degrees, ± 60 degrees, or ± 30 degrees. For example, the operator seat **140** may have a range of rotation of approximately ± 15 - 25 degrees from a forward-facing direction, ± 20 - 25 degrees from a forward-facing direction, or ± 18 - 22 degrees from a forward-facing direction. In one embodiment, the operator seat **140** has a range of rotation of approximately ± 22 degrees from a forward-facing direction. One will also appreciate that the range of rotation need not be the same in the clockwise and counterclockwise directions.

[0096] The operator's seat **140** can be configured to slide backward, sideways, forward, or a combination of such directions, while rotating. The operator's seat **140** can be adjusted vertically. In an embodiment, the operator's seat **140** has a bottom having front, left, right and rear edges. The operator's seat **140** can have one armrest, or it may have two armrests, or it may be free of armrests. In an embodiment, the operator's seat **140** includes at least one armrest.

[0097] The lift truck **100** includes a lift assembly that is used to lift a load. For example, the mast **110** can support a fork carriage (or carriage back rest) **160** having at least one fork **165** attached thereto. The fork carriage **160** can be raised to different heights by movement of the mast **110** and is part of the lift assembly. The lift assembly is configured to vertically lift a load on the forks using, for example, a hydraulic system. A lift assembly may also include paper roll clamps, carton clamps, multiple-forks assemblies, side-shifters, and other suitable attachments. In an embodiment, the lift assembly includes a reach assembly, such as a pantograph-type reach assembly. The lift assembly is attached to the chassis **130**. The energy source of the lift truck **100** includes components used for the lift assembly, such as a hydraulic tank, a hydraulic pump that is fluidly connected to the hydraulic tank, a motor that is connected to the hydraulic pump, and at least one hydraulic valve. The hydraulic pump may be connected to the hydraulic tank via a hose, or it may be directly connected without a hose.

[0098] In an embodiment, the lift truck **100** includes a tilt mechanism, connected to the lift assembly, to tilt the top of the mast **110** back slightly, and therefore the ends of the forks **165** up slightly, to more stably carry a load.

[0099] The lift truck **100** can include a roof that employs an overhead guard **170** that extends over the operator compartment **105** to protect the operator from hazards, such as falling materials. The lift truck **100** can include a roof over the operator compartment **105** that forms a cover for the operator compartment **105**. In an embodiment, the overhead guard **170** is solid and opaque, for example, to act as a sun shade to protect the operator from sunlight when the truck is used outside. Alternatively, the overhead guard **170** is transparent or semi-transparent in whole or in part and/or has one or more apertures to enable the operator to view the forks **165**, top of the mast **110**, and/or its load when the mast **110** is raised. In another embodiment, the overhead guard **170** comprises a sheet of plexiglass over substantially the entire roof area of the overhead guard **170** so as to maximize overhead visibility for the operator. In an embodiment, a tilt mechanism can be incorporated within, or attached to, the overhead guard.

[0100] In one embodiment, the overhead guard **170** may have an optional notch (not shown) on one side. The notch, when present, may facilitate having a crane, lift, or the like access heavy items within either the operator compartment **105** or a chassis compartment underneath the operator compartment **105**. For example, lead-acid batteries may be hoisted into or out of the chassis compartment using a crane, lift, or the like as part of a battery-replacement operation.

[0101] FIG. 2 illustrates a rear isometric view of the lift truck **100** shown in FIG. 1, having a steering wheel **135** in the operator compartment **105**. Lift truck **100** has two front wheels **192** and two rear wheels **191**. The steering wheel **135** is configured to turn the rear wheels **191** when the lift truck moves forward or backward. The steering wheel **135** is mounted on a steering column **137**. In an embodiment, the steering wheel **135** may be mounted on a steering column **137** that is attached to and extends from the floor **150** of the operator compartment **105**. In an embodiment, there is no steering wheel or steering column present in the operator compartment **105**. For example, the steering controls can be integrated into the operator's seat **140** of the lift truck **100**.

[0102] In reference to FIG. 2, truck **100** has a panel **111** that is located behind the operator's seat **140**. The panel **111** can be a cover to a rear compartment that lies above counterweight **112**, and that may contain, for example, power electronics or other truck parts. The panel **111** is preferably sloped downwardly behind the operator's seat, toward the rear of the truck **100**. The downward slope of the panel **111** enhances operator visibility when the truck **100** travels in a reverse or backward direction. The panel **111** may be a cover for an operator-accessible compartment. In an embodiment, the panel **111** does not act as a cover for a compartment.

[0103] Foot controls such as foot pedals **155**, can be present on the floor **150** of the operator compartment **105**. Foot controls can include an accelerator, a brake, and an inching pedal. Alternatively, the operator compartment floor **150** may be free of foot controls. In an embodiment, there are no foot controls in the operator compartment **105**.

[0104] FIGS. 3 and 4 illustrate side views of the embodiment of the forklift truck **100** shown in FIGS. 1 and 2. In reference to FIGS. 3 and 4, the operator compartment floor **150** is located above the ground **190** at a floor height **195**. The ground **190** is considered to be the surface upon which the lift truck lies and may be a ramp or other raised supporting surface as well as a warehouse floor, a road, or the like. The access to and alternative egress from the operator compartment **105** as well as the operator compartment floor height **195** meet the standards of SAE J185: Recommended Practice for Access Systems for Off-Road Machines.

[0105] The operator compartment floor **150** may be approximately 10%-30% lower than the floor of commonly available lift trucks with similar lifting capacity. In some embodiments, the floor height **195** may be the height of a portion of the operator compartment floor **150**, the height of the majority of the area of the operator compartment floor **150**, or an average height of the operator compartment floor **150**. In some embodiments, the floor height **195** may additionally or alternatively be the lowest point on the operator compartment floor **150**.

[0106] The floor height **195** may additionally or alternatively be defined with respect to other features of the lift trucks **100**, **300**, **500**, **600**, **700**, **800**, and **900** (generically lift truck **100**) in the various drawings. For example, with reference to FIG. 10C (used for convenience of adding height lines), the lift trucks **100** (**300**) have drive wheels such as front wheels **192** that have a drive wheel height **198** and a drive wheel radius **194** that determines a drive wheel radius height **196**. The drive wheel radius height **196** and the floor height **195** also determine a floor-radius differential height **197**. Similarly, the lift trucks **100** (**300**) have steer wheels such as rear wheels **191** that have a steer wheel height **203** and a steer wheel radius **199** that determines a steer wheel radius height **201**. Other truck features useful as landmarks may include a bottom **207**, which may be the bottom of the chassis **130** or the bottom of a component attached to the bottom of the chassis **130**. The bottom **207** may define a ground clearance height **205**.

[0107] In some embodiments, the top surface of the operator compartment floor **150** may be 20 inches (or about 51 cm), or closer, from the ground or surface that supports the forklift truck **100**. Generally, the height **195** of the operator compartment floor **150** from the ground **190** is between approximately 10 and 30 inches (or about 25-76 cm), such as between approximately 15 and 25 inches (or about 38-64 cm), between approximately 17 and 21 inches (or about 43-54 cm), between approximately 18 and 20 inches (or about 45-51 cm), or approximately 19 inches (or about 48 cm).

[0108] In some embodiments, the operator compartment floor **150** has a floor height **195** that is less than or equal to the drive wheel height **198**. The operator compartment floor **150** may have a floor height **195** that is less than or equal to 95% of the drive wheel height **198**, less than or equal to 90% the drive wheel height **198**, less than or equal to 85% the drive wheel height **198**, less than or equal to 80% the drive wheel height **198**, less than or equal to 75% the drive wheel height **198**. In some embodiments, the operator compartment floor **150** has a floor height **195** that is between the drive wheel height **198** and the drive wheel radius height **196**. In some embodiments, the operator compartment floor **150** has a floor height **195** that is between the drive wheel height **198** and the steer wheel radius height **201**.

[0109] The term “low floor,” as used herein, refers to an operator compartment floor having a height from the ground that can be comfortably reached by a typical (e.g., “95%,” meaning all but the 5% outliers of the population in terms of size and shape) male or female truck operator using only a single vertical stepping motion. That is, there may be no steps intermediate the operator compartment floor **150** and the ground **190**. Although use of a grab bar may not be necessary by an operator to comfortably enter or exit the operator compartment **105**, a grab bar (not shown) may nonetheless be located on the truck **100**, such as attached to a front strut **188** of the chassis **130**, to reinforce operator training for entering and exiting the truck **100**. In an embodiment, the height **195** from the ground **190** does not require 95% of male or female operators to raise (or extend) their leg more than approximately 90 degrees at their hip from standing, while their knee is bent (or flexed) at no more than approximately 90 degrees.

[0110] An operator can enter the lift truck **100** having a low floor **150** with a single step, then pivot to sit in the operator's seat **140**. An operator sitting in the seat **140**, which is preferably rotatable, can rotate the seat about vertical axis **145** to view in front of, to either side of, and behind the lift truck **100**, with minimal twisting or neck or back strain. Embodiments of the operator's seat **140** can be designed to slide forward, sideways, backward, or a combination of such directions, independent from, or concurrent with, rotation. Thus, for example, the operator's seat may slide backwards and sideways upon rotation to enable the operator's legs and feet to be laterally raised above the floor **150** yet maintained within the operator compartment **105** while rotating.

[0111] The operator's seat **140** may include a mechanism that adjusts the seat vertically, in addition to or instead of the rotation and/or sliding mechanisms. For example, embodiments of the operator's seat **140** can be designed to go up and down from an initial height between approximately 0.5 and 10 inches (or about 1-26 cm), such as between approximately 1 and 6 inches (or about 2-16 cm), between approximately 2 and 5 inches (or about 5-13 cm), at least approximately 2 inches (5 cm), no more than approximately 5 inches (13 cm), or approximately 4 inches (10 cm).

[0112] In contrast to commonly available lift truck floors, the low floor **150** provides adequate leg room for the operator's legs to comfortably rotate about vertical axis **145** while maintaining their height above the floor, when the operator is sitting in the operator's seat **140**. For lift trucks having a seat **140** that can slide backward, or backward and sideways, while rotating more than a pre-determined angular displacement, for example, more than **10** degrees, the operator's legs can be maintained within the operator compartment **105** as the seat **140** rotates. In an embodiment, there is an amount of unobstructed leg room on the floor **150** of the operator compartment **105** to permit an operator of the lift truck **100** to swing their legs laterally while keeping their feet above the floor **150**.

[0113] The chassis compartment may have a chassis compartment area between the one or more drive wheels and the counterweight **112** (or between the drive axle **324** and the counterweight **112**) and between the left side and right side of the chassis **130**. In some embodiments, the operator compartment floor **150** has a floor area that is greater than or equal to 95% of the chassis compartment top area, greater than or equal to 90% of the chassis compartment top area, greater than or equal to 85% of the chassis compartment top area, greater than or equal to 80% of the chassis compartment top area, greater than or equal to 75% of the chassis compartment top area, or greater than or equal to 60% of the chassis top compartment area. As used herein, the chassis compartment top area is the area of the chassis compartment on its top side.

[0114] In some embodiments, the operator compartment **105** has an operator compartment area between the drive axle **324** and the counterweight **112** and between the left side and right side of the chassis **130**, and the operator compartment floor **150** has a floor area that is greater than or equal to 95%, 90%, 85%, 80%, 75%, or 60% of the operator compartment area.

[0115] In some embodiments, the lift truck includes an overhead guard **170** that has an overhead-guard area between front struts **188** and rear struts that may support the overhead guard **170**, and

the operator compartment floor may have a floor area that is greater than or equal to the overhead-guard area, or 95%, 90%, 85%, 80%, 75%, or 60% of the overhead-guard area.

[0116] FIG. 32 is a top view of a lift truck similar to the lift truck 100 in FIG. 3, comparing conventional floor surface area 161 (where conventional foot controls are located) of a conventional lift truck to a total floor surface area 163 of the operator compartment floor 150, wherein the total floor surface area 163 equals the conventional floor surface area 161 plus gained floor surface area 167 (including area under and to the sides of the operator seat 140), according to one embodiment of a lift truck 100 disclosed herein. The gained floor surface area 167 may be as large as three times larger than the conventional floor surface area 161. In some embodiments, the total floor surface area 163 may be as large as four times larger than the conventional floor surface area 161. In some embodiments, the total floor surface area 163 may be greater than or equal to three times larger than the conventional floor surface area 161.

[0117] The lift truck 100 includes a plurality of wheels, such as two rear wheels 191 and two front wheels 192. The wheels 191 and 192 are directly or indirectly attached to the chassis 130 of the truck 100 and are configured to roll to steerably move the truck 100 forward and backward. For example, rear wheels 191 may be attached to a steer axle 189 that is attached to a counterweight 112, which is attached to chassis 130, and front wheels 192 may be attached to a drive axle 324 (FIG. 10B) that is attached to chassis 130. All or a subset of the plurality of wheels may be steerable by an operator.

[0118] The lift truck 100 generally has two pairs of wheels, as indicated in FIGS. 1-4. Two of the wheels 191 and 192 of the lift truck 100 can independently be replaced with a single wheel to form a lift truck having three wheels. For example, the two rear wheels 191 shown in FIG. 2 may be replaced with a single rear wheel. Additional front or rear wheels can independently be added to the lift truck 100 to form a truck having more than four wheels. In an embodiment, the lift truck 100 has a plurality of wheels that comprises at least one left wheel and at least one right wheel. In an embodiment, the lift truck 100 has a plurality of wheels that comprises two front wheels and one rear wheel. All or a subset of the wheels may be solid, or all or a subset of the wheels may be pneumatic, or they may include a mixture of solid and pneumatic wheels.

[0119] FIG. 5 is a top view of the lift truck 100 shown in FIGS. 1-4. The lift truck chassis 130 (not shown in FIG. 5) includes a section in front of, or anterior to, the operator's seat 140 and a section that is behind, or posterior to, the seat 140. A plane 200 can include the vertical axis 145 (not shown in FIG. 5) of the seat 140, extending from the vertical axis 145 and be orthogonal to the right and left sides of the truck 100. The plane 200 is orthogonal to the longitudinal axis 125 and essentially divides the truck chassis 130 into two portions. The area of the truck 100 that generally lies in front of the plane 200 and behind the forks 165 is referred to herein as the anterior portion 210, as indicated in FIG. 5. Similarly, the area of the truck 100 that is generally behind the plane 200 is referred to herein as the posterior portion 220.

[0120] The anterior portion 210 of the truck 100 generally includes components of the lift assembly, as well as means for steering and the front wheels. The posterior portion 220 can include at least one counterweight 112 and the rear wheel or wheels 191. The operator compartment 105, operator compartment floor 150, overhead guard, and components of the energy source are generally located in both the anterior portion 210 and the posterior portion 220, such as within chassis compartments that straddle or overlap both portions 210 and 220.

[0121] FIGS. 6, 7A, 7B, 8 and 9 illustrate an embodiment of a lift truck 300 having a low floor 350, a chassis 330, a counterweight 312, a rear compartment 313, and neither a steering column nor a steering wheel. Means for steering truck 300 are hand-operated controls that are integrated into an operator seat 340, and such means are referred to herein as "seatside steering." The seat 340 can incorporate steering controls, such as by integration of hand-operated operator controls into the seat 340. Such integrated steering controls may be located on an armrest, or on both armrests, of the seat. The seat 340 can preferably rotate about a vertical axis 345.

[0122] Seat **340** is shown attached to the truck **300** by a seat bracket **341** that extends from the truck chassis **330** in the posterior portion of the chassis **220** (FIG. 5). The seat **340** can alternatively be attached to the truck **300** by a bracket that extends from the floor **350**, such as a column that is parallel to, or coincident with, the vertical axis of rotation **345**. Truck **300** also includes an overhead guard **370**, two rear wheels **391** and two front wheels **392**, an operator compartment **305**, a lift assembly **360** having forks **365**, mast **310**, and a lift cylinder **315** (FIG. 11). Unless specifically indicated otherwise, embodiments of the truck **300** can include some or all of the various components and variations thereof, as described for truck **100** above, such as an optional notch in the overhead guard **370**.

[0123] As shown in detail in FIGS. 7A and 7B, the truck **300** has a steering knob **331** incorporated into an armrest of the operator's seat **340** and foot pedals **355** extending from, and close to, the operator compartment floor **350**. In an embodiment, the operator can control the speed of the truck using foot controls and steer the truck using seatside steering. Exemplary controls used as means for seatside steering can include a steering knob that steers the truck, and finger controls that actuate the lift assembly. Controls used for actuating the lift assembly may be integrated wholly or in part into the operator's seat, such as into an armrest, or may include foot controls in addition to, or instead of, hand-operated controls.

[0124] The truck chassis **330** can have sides and a bottom that are configured to define a chassis compartment within the sides and above the bottom. The truck **300** includes an operator compartment **305** having a low operator compartment floor **350**. Low floor **350** can be formed from a plate that is secured to the truck chassis **330**, such as by welding or via screws, bolts, or the like. The low floor **350** can be wholly or partially formed from a plate. The operator compartment **305**, in an embodiment, can be bounded on its bottom by a plate on, and secured to the chassis **330**, with the plate forming the floor **350**. Such a plate can form the top cover of a chassis compartment that is located underneath seat **340**, upon which an operator's feet can rest when the operator is seated. The plate can provide structural strength to the chassis **330**, such as to resist deflection of the chassis, in addition to forming a cover for a chassis compartment.

[0125] The operator compartment floor **350** may be substantially flat in one or more directions, such as side-to-side, with reference to an operator seated in the seat **340** having the vertical axis **345** and looking straight toward the front of the truck along a longitudinal axis **325**. As used herein, the term “substantially” is used herein as a broadening term to refer to an amount at least about 75%, such as about 80%, about 85%, about 90%, about 95%, about 99%, or 100% of an object or characteristic. A floor **350** that is substantially flat side-to-side is therefore at least about 75% flat, and may be perfectly flat, in the left and right side directions as the operator looks forward from the seat **340**. In some embodiments, the operator compartment floor **350** has a floor area that is substantially flat for greater than or equal to 50% of the chassis compartment area, that is substantially flat for greater than or equal to 60% of the chassis compartment area, or that is substantially flat for greater than or equal to 70% of the chassis compartment area.

[0126] In an embodiment, the plate may be substantially flat side-to-side, but the floor **350** may have portions that are sloped in one or more directions front-to-back. For example, a portion of the floor **350** that is immediately underneath the operator's seat **340** may be substantially flat, but a portion of the floor that is anterior to the seat **340** and behind a fork carriage **360** may slope upward toward the fork carriage **360** and may accommodate foot controls such as foot pedals **355**. In an embodiment, the floor **350** extends substantially flat front-to-back from a front cowl that is proximate to the lift truck, to a rear floor edge that is behind the front edge of the seat **340**.

[0127] Embodiments of a lift truck can include a compartment that is readily accessible to allow for easy access to a specific component. Optionally, such a readily accessible compartment may be included as part of the chassis or may be included as part of the counterweight. Referring to FIGS. 7A and 7B, a compartment **313** can be located in the posterior portion of the truck **300**, above the counterweight **312** and behind the seat **340**. The compartment **313** can include a cover that is

formed by panel **311** that can be readily opened by an operator (see FIG. 7B), and therefore can be an operator-accessible compartment. For example, the cover formed by panel **311** may be secured in place via friction in cover hinges, may have a snap-fit securement, or other suitable closure that can be opened by an operator without the use of tools.

[0128] In an embodiment, the truck **300** can have an operator-accessible compartment **313** that contains at least one operator-serviceable component **314**. The operator-serviceable components may include power electronics, such as one or more motor controllers, fuses, VSM, contactors or other suitable operator-serviceable components. An operator-accessible compartment **313** can be integrated into and/or above the counterweight **312** of the truck **300**, such as via a plate attached to the top of the counterweight **312** that forms a cover using the panel **311** for the chassis compartment **313**. Operator-serviceable components may however be located in other places in the truck, such as underneath the seat **340**, for example.

[0129] Because the compartment **313** can be located behind the seat **340**, the top cover made by the panel **311** of the compartment **313** can potentially hinder the operator's view behind the truck **300**. It can be useful, therefore, to design the compartment **313** and the panel **311** such that they do not adversely impact operator visibility. In an embodiment, the panel **311** that covers compartment **313** slopes downward toward the rear of the lift truck **300**, as indicated in FIGS. 7A and 7B. This downward slope creates a clear sight line over the panel **311** for the operator when sitting in the operator seat **340** and permits the operator to see low objects immediately behind the lift truck **300**.

[0130] This sight-line advantage is demonstrated in FIG. 31, which compares a conventional sight line **317** of an old lift truck to a sight line **319** achievable according to one embodiment of the lift trucks **100**, **300**, **500**, **600**, **700** and **800** disclosed herein. With reference to FIG. 31, an operator of average male height seated in the operator seat **340** may have an eye position or eye height **321** that is about 812.8 mm above the cushion of the operator seat **340**. In a conventional lift truck, this operator with a sight line **317** will not be able to see an object such as a cubic block **323** having a block height **327** of 165 mm (such as the height of a short pallet) within a rear distance **329** of 892 mm from the lift truck. However, in embodiments of a lift truck **100**, **300**, **500**, **600**, **700** and **800** disclosed herein, the operator with the improved sight line **319** will be able to see the block **323** within 892 mm behind the lift truck. Moreover, the operator with the improved sight line **319** will be able to see the block **323** within a rear distance **333** within 388 mm behind the lift truck. This sight line advantage provides an operator with a field of view that has a 10 degree or better angle **337** advantage over that permitted in conventional lift trucks. This sight-line advantage also translates in terms of an operator being able to see objects 0.6 meters closer to the rearmost point of the vehicle and effectively reduces the rear blind spot length by about 50%.

[0131] Thus, in some embodiments, the lift truck **300** provides an unobstructed direct sight line **319** from an operator of average male height seated in the operator seat **340** to a front top of an object less than or equal to 165 cm high within 850 mm behind the lift truck **300**. The lift truck **300** may provide an unobstructed direct sight line **319** from an operator of average male height seated in the operator seat **340** to a front top of an object less than or equal to 165 mm high within 800 mm behind the lift truck **300**. The lift truck **300** may provide an unobstructed direct sight line **319** from an operator of average male height seated in the operator seat **340** to a front top of an object less than or equal to 165 mm high within 700 mm behind the lift truck **300**. The lift truck **300** may provide an unobstructed direct sight line **319** from an operator of average male height seated in the operator seat **340** to a front top of an object less than or equal to 165 mm high within 600 mm behind the lift truck **300**. The lift truck **300** may provide an unobstructed direct sight line **319** from an operator of average male height seated in the operator seat **340** to a front top of an object less than or equal to 165 mm high within 500 mm behind the lift truck **300**. The lift truck **300** may provide an unobstructed direct sight line **319** from an operator of average male height seated in the operator seat **340** to a front top of an object less than or equal to 165 mm high within 400 mm behind the lift truck **300**. The lift truck **300** may provide an unobstructed direct sight line **319** from

an operator of average male height seated in the operator seat **340** to a front top of an object less than or equal to 165 mm high within 388 mm behind the lift truck **300**.

[0132] FIGS. **10A**, **10B**, **10C** and **10D** illustrate embodiments of a lift truck **300** having hand-operated controls that are integrated into an operator seat **340** and that is free from pedals or foot controls, and also free from a steering wheel and steering column. For clarity, the lift carriage and mast components are removed from these figures. In the embodiments shown, the seat **340** integrates hand-operated operator controls into the armrests of the seat **340**. These hand-operator controls include a steering knob **331** integrated into one armrest and a set of finger controls **332** integrated into the second armrest. In an embodiment, the hand-operator controls include only one of knob **331** and finger controls **332**. The finger controls **332** can, for example, actuate the lift carriage and control the speed of the truck **300**. The steering knob **331** can steer the truck.

[0133] Referring to FIGS. **10A** and **10B**, embodiments of lift truck **300** can include a front cowl **353**. The cowl **353** can be made of metal or plastic, or a combination thereof. The cowl **353** may be partially or completely transparent, to allow for an operator to view the lift carriage from the seat **340**. In an embodiment, the cowl **353** is made of plexiglass. The cowl **353** may include a shelf or other means for storage on the inside of the operator compartment **305**, such as a cup holder or a holder for a computerized device, such as a GPS device or an inventory device.

[0134] Referring to FIG. **10D**, which provides a close-up view to show greater details, embodiments of lift truck **300** can include a seat **340** having finger controls **332** that are placed on the underside of an armrest, which also has a wrist strap.

[0135] FIG. **11** is a front view of a lift truck **300**. In the embodiment of the lift truck **300** illustrated in FIG. **11** having a substantially flat and low floor **350**, the operator has good visibility for counterbalance-type trucks in front of the truck because the components in the anterior portion of the truck **300** are lower in height as compared to components in commonly available lift trucks. The low floor **350** allows, for example, the components of the lift assembly that lie behind the fork carriage **360** and in front of and underneath the seat **340**, which may include a hydraulic system used to raise and to lower the forks **365**, to be placed lower within the chassis **330** than those of commonly available trucks. In an embodiment, the cowl **353** is transparent.

[0136] Having the low floor can also provide additional advantages, such as providing a lift truck **300** with a lower center of gravity than commonly available trucks because many of the heavy components of the truck **300** are located closer to the ground and/or in a horizontal arrangement rather than a vertical arrangement. The lift truck **300** may provide a center of gravity situated at a gravitational center height that is lower than or equal to the drive wheel height **198**. The lift truck **300** may provide a center of gravity situated at a gravitational center height that is lower than or equal to the floor height **195**. The lift truck **300** may provide a center of gravity situated at a gravitational center height that is lower than or equal to the steer wheel height **203**. The lift truck **300** may provide a center of gravity situated at a gravitational center height that is lower than or equal to the drive wheel radius height **196**. The lift truck **300** may provide a center of gravity situated at a gravitational center height that is lower than or equal to the steer wheel radius height **201**.

[0137] FIG. **12** is a top view of the truck **300** and illustrates the broad and unobstructed low floor **350** of the operator compartment **305**. Seatside steering controls include finger controls **332** and a steering knob **331**, which integrate into an armrest of the seat **340**. Thus, no steering wheel or steering column is needed in this embodiment. The floor **350** of the operator compartment **305** can extend substantially flat side-to-side across approximately the entire width of the lift truck **300**. When an operator is seated, there are no obstructions on the floor **350** that can hinder the operator's feet and legs when rotating the seat **340** to, for example, look around the truck **300** while driving or to exit the truck **300**. In an embodiment, the lift truck **300** includes a steering wheel that may be mounted to a steering column that does not extend from the floor **350**. For example, the steering column may be attached to or integrated with the seat **340**, and as such, does not impede the

movement of the operator's legs or feet.

[0138] In some embodiments, the truck **300** has seatside steering, such as a steering knob **331** and/or finger controls **332**, and one or more foot controls in the operator compartment **305** that are near the floor **350** for operation by the driver. Exemplary foot controls include pedals. In an embodiment, the lift truck **300** includes an operator's seat **340** that has a bottom having front, left, right and rear edges, and a floor **350** that extends substantially flat front-to-back from a front floor edge proximate the front of the forklift truck **300** to a rear floor edge behind the front edge of the bottom of the seat **340**.

[0139] FIGS. **13** and **14** illustrate an embodiment of a lift truck **500** having an exemplary common truck chassis **530** that can be used for a lift truck having seatside steering and a lithium-ion battery **520** as an energy source. FIG. **13** shows the chassis **530** with the operator compartment **505** removed, to more clearly illustrate a chassis compartment under the floor of the operator compartment **505**. Referring to FIG. **13**, a chassis compartment is formed from chassis sides **501** and a bottom (not shown), to form a compartment that is within the sides and above the bottom of the chassis **530**. The chassis compartment can contain, for example, an energy source including a lithium-ion battery **520** and an electric motor. A hydraulic system **580** may also be included in the chassis compartment. As illustrated in FIGS. **13** and **14**, embodiments of the chassis compartment can extend above the front wheels in addition to under the operator's seat, as well as behind the operator's seat. In FIG. **14**, for example, a counterweight **512** and additional truck parts **514** are shown attached behind the seat **540** in (wholly or partially) or on a chassis compartment on the rear side of the truck **500**.

[0140] An exploded view of the truck **500** is shown in FIG. **14**, which shows the operator compartment **505** elevated above lithium-ion battery **520**, which is shown elevated above the chassis **530** and other components. FIG. **14** shows the chassis **530** having a bottom **502** of a chassis compartment and sides **501**, a panel **511** (which can be sloped or level), a counterweight **512** and the hydraulic system **580**.

[0141] The counterweight **512** may be made of one or more pieces, with each piece having one or more densities, such as a two-piece counterweight comprising multiple SAE J431 Gray Iron Casting grades. The truck **500** also includes an overhead guard **570**, a lift assembly **560** having forks **365**, mast **510**, and hand-operated controls **531** and **532** integrated into seat **540**. Unless specifically indicated otherwise, embodiments of the truck **500** can include some or all of the various components and variations thereof, as described for trucks **100** and **300** above, such as an optional notch in the overhead guard **570**.

[0142] FIGS. **15A** and **15B** are top views of embodiments of the truck **500** with the operator compartment **505** removed, showing the battery bank and chassis compartments. FIGS. **15A** and **15B** show the lithium-ion battery **520** and the hydraulic system **580** in more detail. The smaller size of a lithium-ion battery **520** as compared to a lead-acid battery can provide an energy source with a low profile. A lift truck **500** can be designed to contain a low-profile lithium-ion battery **520** that is placed in a chassis compartment underneath the operator's seat **540**, with a counterweight **512** placed in the rear of the truck **500** and behind the operator's seat **540**. Using smaller lithium-ion batteries **520** for the energy source instead of larger, bulkier lead-acid batteries permits the chassis compartment containing the energy source to be smaller, therefore allowing for the low operator compartment floor **550**, while maintaining similar amounts of electrical energy typically stored by lead-acid batteries in commonly available fork lift trucks.

[0143] The lithium-ion battery bank **520** is shown within a chassis compartment underneath the seat **540**, behind (posterior to) most of the hydraulic system **580**, and in front of (anterior to) counterweight **512** and truck parts **514**. The arrangement of the components that make up the hydraulic system **580** and energy source may vary, in some embodiments of the lift trucks disclosed herein, in a vertical and/or horizontal direction as compared to the embodiment shown in FIGS. **13-15**.

[0144] The hydraulic system **580** includes a hydraulic motor **581**, a hydraulic tank **582**, a hydraulic pump **583**, at least one hydraulic valve **584**, and at least one hydraulic cylinder **585**, such as a tilt cylinder. The hydraulic pump **583** is connected to the hydraulic tank **582** such that the fluid in the tank **582** can be pressurized and released when the pressure in the hydraulic pump **583** is increased and decreased, respectively. The connection can be direct or indirect. For example, the hydraulic pump **583** can be directly connected to the hydraulic tank **582** via a plastic, metallic, or other appropriate hose. In an embodiment, the hydraulic pump **583** is directly connected to the hydraulic tank **582** without a hose. In one embodiment, the hydraulic motor **581** includes hydraulic lift cylinders.

[0145] The hydraulic pump **583** and hydraulic motor **581** may be located in a chassis compartment. The hydraulic tank **582** is fluidly connected to the hydraulic pump **583**. The hydraulic tank **582** may be located in the same chassis compartment as the pump **583** and/or the motor **581**, or the hydraulic tank **582** may be located in a different chassis compartment.

[0146] The hydraulic system **580**, in an embodiment, includes one hydraulic valve **584** that controls the mast functions, and a different hydraulic valve **584** that controls the tilt functions for the mast **510**. In one embodiment, the drive axle **524** is located at the front of the lift truck **500** and is attached to a front side of the chassis **530**. In an embodiment, the drive axle **524** may form part of a drive train and can include one or more electric motors **525**. In one embodiment, one or more electric motors **525** may be located in the drive axle **524**. The drive axle may be attached to the mast **510**.

[0147] Using the lithium-ion battery **520**, the wheel-turning electric motor and the hydraulic system **580** constitute a power plant for the lift truck **500**. The power plant drives at least some of the wheels to move the lift truck **500** and also raises the mast **510** to lift loads. Different embodiments, such as those described below, have different power plants based on different energy sources in place of the lithium-ion battery **520**, but the lithium-ion battery **520** is presently a preferred energy source because it provides sufficient energy to use the lift truck during a complete work shift between recharging in a small volume that enables the size of the chassis compartment to be reduced and the floor of the operator compartment **505** to therefore be lowered, thereby realizing ergonomic advantages for the operator. A lithium-ion battery **520** may include, for example, multiple lithium cells connected together to achieve desired voltage and current for the lift truck **500** and/or a battery management system and battery power delivery components. Lithium-based batteries may include lithium-ion batteries as well as other batteries that include lithium in their chemical compositions. Lithium-ion batteries **520** may include chemical compositions of one or more of: lithium titanate, lithium-iron phosphate, lithium-nickel manganese cobalt oxide, lithium-manganese oxide. Other lithium battery chemistries that do not fall in the category of lithium ion include lithium sulfur, lithium air, and lithium silicone. One will appreciate that other types of lithium-based battery may be used in place of a lithium-ion battery **520** if the particular lithium-based battery has properties suitable for lift truck applications.

[0148] FIGS. **16** and **17** illustrate an embodiment of a lift truck **600** having a chassis **630**, seatside steering and using a lead-acid battery as the energy source. Truck **600** also includes an overhead guard **670**, a lift chassis **660** having forks **665**, mast **610**, and hand-operated controls **631** and **632** integrated into seat **640**. The seat **640** can preferably rotate about a vertical axis **645**. Unless specifically indicated otherwise, embodiments of truck **600** can include some or all of the various components and variations thereof, as described for trucks **100**, **300**, and **500** above, such as an optional notch in the overhead guard **670**. Referring to FIG. **16**, the floor **650** of the operator compartment **605** is not a low floor, in order to accommodate the larger size of a currently available lead-acid battery bank **620** underneath the operator's seat **640**, as compared to the low-profile and smaller lithium-ion battery bank **520** shown for the truck **500** in FIGS. **13-15**.

[0149] An exploded view of the truck **600** is shown in FIG. **17**, which shows the operator compartment **605** elevated above the lead-acid battery bank **620**, which is shown elevated above

the chassis **630**. In the embodiment shown in FIGS. **16** and **17**, the floor **650** of the operator compartment **605** has neither a steering column nor a steering wheel. FIG. **17** shows the chassis **630** having a bottom **602** of a chassis compartment and sides **601**. The hydraulic system **680** can fit completely or partially within the chassis compartment.

[0150] Embodiments of the truck **600** can contain a counterweight **612** that may be made of one or more pieces, with each piece having one or more densities, such as a two-piece counterweight comprising multiple SAE J431 Gray Iron Casting grades. The counterweight **612** may be located behind the operator's seat **640**.

[0151] FIGS. **18A** and **18B** are top views of embodiments of the truck **600** with the operator compartment **605** removed, showing the battery bank and chassis compartments. FIGS. **18A** and **18B** show the lead-acid battery bank **620** and the hydraulic system **680** in more detail. The lead-acid battery bank **620** is shown within a chassis compartment underneath the seat **640**, behind (posterior to) most of the hydraulic system **680**, and in front of (anterior to) counterweight **612** and truck parts **614**. Unless specifically indicated otherwise, the energy source, hydraulic system and drive mechanisms of embodiments of truck **600** can include some or all of the various components and variations thereof, as described for trucks **100**, **300** and **500** above.

[0152] The hydraulic system **680** includes a hydraulic motor **681**, a hydraulic tank **682**, a hydraulic pump **683**, at least one hydraulic valve **684**, and at least one hydraulic cylinder **685**. The functions and construction of the hydraulic system **680** and its components may be as described above in relation to the hydraulic system **580**. In one embodiment, the drive axle **624** is located at the front of the lift truck **600** and is attached to a front side of the chassis **630**. In an embodiment, the drive axle **624** may form part of a drive train and can include one or more electric motors **625**.

[0153] FIGS. **19** and **20** illustrate an embodiment of a lift truck **700** having a chassis **730**, seatside steering and using a hydrogen fuel cell **720** as part of the energy source. The associated power plant may include an electric motor, power electronics, a hydrogen delivery system, and power connectors. Truck **700** also includes an overhead guard **770**, a lift assembly **760** having forks **765**, mast **710**, and hand-operated controls **731** and **732** integrated into seat **740**. Unless specifically indicated otherwise, embodiments of truck **700** can include some or all of the various components and variations thereof, as described for trucks **100**, **300**, **500**, and **600** above, such as an optional notch in the overhead guard **770**. Referring to FIG. **19**, the floor **750** of the operator compartment **705** is not a low floor, to accommodate the larger size of the hydrogen fuel cell components underneath the operator's seat **740**, as compared to the lower-profile and smaller lithium-ion battery bank **520** shown in the truck **500** of FIGS. **13-15**.

[0154] An exploded view of the truck **700** is shown in FIG. **20**, which shows the operator compartment **705** elevated above the energy source (fuel cell power system or fuel cell power plant) **720** which is shown elevated above the chassis **730**. In some embodiments, the fuel cell power system **720** includes a hydrogen fuel cell **721**, a cooling system **723**, and fuel cell engine or motor **727**. In the embodiment shown in FIGS. **19** and **20**, the floor **750** of the operator compartment **705** has no steering column nor steering wheel. FIG. **20** shows the chassis **730** having a bottom **702** of a chassis compartment and sides **701**. The hydraulic system **780** may fit completely or partially in the chassis compartment.

[0155] Embodiments of the truck **700** can contain a counterweight **712** that may be made of one or more pieces, with each piece having one or more densities, such as a two-piece counterweight comprising multiple SAE J431 Gray Iron Casting grades. The counterweight **712** may be located behind the operator's seat **740** and underneath hydrogen storage tanks used for the hydrogen fuel cell **721**. FIG. **20** shows the truck **700** having a rear chassis compartment also behind the operator's seat **740**, the compartment covered by panel **711**, with the hydrogen storage tanks resting above panel **711**.

[0156] FIGS. **21A** and **21B** are top views of embodiments of the truck **700** with the operator compartment **705** removed, showing the hydrogen-based energy source **720** (such as a hydrogen

fuel cell power system or hydrogen fuel cell power plant), hydrogen storage tanks **729**, and chassis compartments. FIGS. **21A** and **21B** show the energy source **720** and the hydraulic system **780** in more detail. Most of the hydrogen fuel cell power plant **720** is shown within a chassis compartment underneath the seat **740**, behind (posterior to) most of the hydraulic system **780**, and in front of (anterior to) counterweight **712** and truck parts **714**. Unless specifically indicated otherwise, the energy source, hydraulic system and drive mechanisms of embodiments of truck **700** can include some or all of the various components and variations thereof, as described for trucks **100**, **300**, **500** and **600** above.

[0157] The hydraulic system **780** includes a hydraulic motor **781**, a hydraulic tank **782**, a hydraulic pump **783**, at least one hydraulic valve **784**, and at least one hydraulic cylinder **785**. The functions and construction of the hydraulic system **780** and its components may be as described above in relation to the hydraulic system **580**. In one embodiment, the drive axle **724** is located at the front of the lift truck **700** and is attached to a front side of the chassis **730**. In an embodiment, the drive axle **724** may form part of a drive train and can include one or more electric motors **725**. The hydrogen fuel cell power plant **720** may provide energy directly to the hydraulic motor **781** or may provide energy to charge batteries **731**, which can be regulated to feed the hydraulic motor **781**.

[0158] FIGS. **22** and **23** illustrate an embodiment of a lift truck **800** having a chassis **830**, seatside steering and using an internal combustion engine as the energy source. The truck **800** also includes an overhead guard **870**, a lift chassis **860** having forks **865**, mast **810**, and hand-operated controls **831** and **832** integrated into seat **840**. Unless specifically indicated otherwise, embodiments of the truck **800** can include some or all of the various components and variations thereof, as described for trucks **100**, **300**, **500**, **600** and **700** above, such as an optional notch in the overhead guard **870**. Referring to FIG. **22**, the floor **850** of the operator compartment **805** is not a low floor, to accommodate the larger size of the internal combustion engine underneath the operator's seat **840**, as compared to the lower-profile and smaller lithium-ion battery shown in the truck **500** of FIGS. **13-15**.

[0159] An exploded view of the truck **800** is shown in FIG. **23**, which shows the operator compartment **805** elevated above internal combustion engine energy source (power plant or drive train) **820**, which is shown elevated above the chassis **830**. The drive train **820** may include a cooling system **823**, an internal combustion engine **827**, and a transmission **833**. In the embodiment shown in FIGS. **22** and **23**, the floor **850** of the operator compartment **805** has no foot controls. FIG. **23** shows the chassis **830** having a bottom **802** of a chassis compartment and sides **801**, and the hydraulic system **880**. The hydraulic system **880** may fit completely or partially in the chassis compartment.

[0160] Embodiments of the truck **800** can contain a counterweight **812** that may be made of one or more pieces, with each piece having one or more densities, such as a two-piece counterweight comprising multiple SAE J431 Gray Iron Casting grades. The counterweight **812** may be placed in a compartment behind the operator's seat **840**. FIG. **23** shows the truck **800** having a rear chassis compartment also behind the operator's seat **840**, the compartment covered by panel **811**, with the portions of the internal combustion engine resting above panel **811**.

[0161] FIGS. **24A** and **24B** are top views of embodiments of the truck **800** with the operator compartment **805** removed, showing the internal combustion-based energy source **820** and chassis compartments. FIGS. **24A** and **24B** show the energy source **820** and the hydraulic system **880** in more detail; however, the illustration of the energy source blocks the view of many of the hydraulic system components. Most of the internal-combustion-based energy source **820** is shown within a chassis compartment underneath the seat **840**, behind (posterior to) most of the hydraulic system **880**, and in front of (anterior to) counterweight **812**. Unless specifically indicated otherwise, the energy source, hydraulic system and drive mechanisms of embodiments of truck **800** can include some or all of the various components and variations thereof, as described for trucks **100**, **300**, **500**, **600** and **700** above.

[0162] The hydraulic system **880** includes a hydraulic motor (not shown), a hydraulic tank **882**, a hydraulic pump (not shown), at least one hydraulic valve **884**, and at least one hydraulic cylinder **885**. The functions and construction of the hydraulic system **880** and its components may be as described above in relation to the hydraulic system **780**. In one embodiment, the drive axle **824** is located at the front of the lift truck **800** and is attached to a front side of the chassis **830**.

[0163] The lift trucks disclosed herein can include a truck chassis that is common to multiple embodiments. For example, the chassis **130**, **330**, **530**, **630**, **730** and **830** may all have the same design, that is, be a common chassis. The common chassis has a chassis cavity or compartment as described and referred to extensively above. An advantage of the common chassis is that the lift truck manufacturer can use the same chassis when manufacturing lift trucks having different floor heights, energy sources, and/or operator compartments. For example, the common chassis can be used in lift trucks that incorporate low floors and also for trucks having floors that are of commonly available height. Similarly, a common chassis can be used in lift trucks having a low-profile battery and also for trucks that use a hydrogen fuel-cell-based energy source.

[0164] Additional aspects of the modular design of the lift trucks and their chassis in the embodiments described herein, are shown in FIGS. **25-30**. The various different designs for the chassis and components that are connected thereto are referred to herein as “form factors.” Parameters contemplated by the form factor include the size, specification, materials used for, and the shape of parts used in the lift truck. Embodiments of the lift trucks shown in FIGS. **25-30** may include various components that can be incorporated in different embodiments of any of the lift trucks disclosed herein. Similarly, components of the lift trucks as described herein for trucks **100**, **300**, **500**, **600**, **700** and **800** above, may be included in any of the embodiments of lift trucks shown in FIGS. **25-30**.

[0165] FIGS. **25A** and **25B** illustrate various shapes of a counterweight system **1000** that can be incorporated into the chassis of embodiments of a lift truck. The counterweight system **1000** can be made with more than one piece, and different form factors can accommodate different shapes and/or arrangements of the lift truck components. For example, different designs for counterweight systems **1000a** and **1000b** (collectively counterweight system **1000**) that are placed behind an operator's seat are shown respectively from the rear left side of a cut away lift truck (FIG. **25A**) and from the front right side of a cut away lift truck (FIG. **25B**).

[0166] With reference to FIGS. **25A** and **25B**, the counterweight systems **1000** may include a structural (or lower) counterweight **1012** that forms part of a rear portion of the chassis **1030** and may be shaped to accommodate various sizes of rear wheels **1091**. The structural counterweight **1012** is typically very heavy and may be the heaviest part of the counterweight system **1000**. The structural counterweight **1012** may be a single piece built all at once, integrally or separately from the chassis **1030**, or the structural counterweight **1012** may be formed of multiple pieces (such as two pieces) connected to each other and/or the chassis **1030** such as by one or more weldments. The structural counterweight **1012** may be adapted to provide attachment points, shapes, or other connector enhancements to support or accommodate the other components of the counterweight system **1000**.

[0167] The counterweight system **1000** may also include a counterweight slab **1014** that rests on top of, and/or fits within, the structural counterweight **1012**. The counterweight slab **1014** is often the second heaviest component of the counterweight system **1000**. The counterweight slab **1014** may form a tray **1016** that is adapted to support a truck component, such as a battery, a fuel tank, an operator accessible component, or an auxiliary counterweight.

[0168] The counterweight system **1000** may also include an overhang counterweight **1018** that may be adapted to hang on or over the counterweight slab **1014** or on or over a truck component supported by the counterweight slab **1014**. A lift truck having a hydrogen-based energy source may, for example, incorporate a counterweight system **1000** that is formed with a tall edge in order to help protect the hydrogen tanks resting thereon. Similarly, a lift truck having a rear chassis

compartment and a lithium-ion battery bank, may incorporate a counterweight system **1000** that is designed to have a shelf or enclosure within it, for an operator-accessible compartment.

[0169] The counterweight system **1000** may also include an upper counterweight **1022** that may be adapted to attach to rear struts of the chassis that support the overhead guard. In some embodiments, the upper counterweight **1022** may function more as an adapter than a weight-bearing component. The design of the upper counterweight **1022** may be coordinated with that of the overhang counterweight **1018** to provide adequate connection between the two components. The upper counterweight **1022** and/or the counterweight system **1000** as a whole may be designed, such as sloped, to enhance the operator sight line to low objects close and behind the lift truck.

[0170] Any of these components of the counterweight system **1000** may be manufactured in multiple weights and/or sizes to suit an interchangeable modular system to accommodate different modular truck configurations utilizing a common chassis **1030** and to fine tune the center of gravity with respect to the particular configuration and specific lift truck components within such configuration.

[0171] FIGS. **26** and **27** are illustrations of an embodiment of a lift truck **900** having seatside steering and a low operator compartment floor **950** having no foot controls, with a different form factor as the lift trucks shown in FIGS. **1-24**. Lift truck **900** has a chassis **930**, seatside steering and a low floor **950**. Truck **900** also includes an overhead guard **970**, a lift chassis **960** having forks **965**, mast **910**, hand-operated controls **931** and **932** integrated into seat **940**, and counterweight **912**. The truck **900** has a lift assembly that includes a tilt mechanism **973** that is incorporated into the overhead guard **970** and is connected to the lift assembly. The tilt mechanism **973** can selectively tilt the lift assembly including fork carriage **960**. The lift truck **900** has a single rear wheel **995**, exemplifying an embodiment of a three-wheeled lift truck.

[0172] In reference to FIGS. **26** and **27**, the low floor **950** is free from any obstruction to the operator's feet or legs when entering or exiting the truck, or when seated in the seat **940**. The operator's seat **940** is attached directly to the operator compartment floor **950** and can rotate about vertical axis **945**. Front cowl **953** of floor **950** separates the fixed operator compartment from the movable lift assembly and provides operator protection.

[0173] The shape and materials used for front cowl **953** may vary. In an embodiment, the cowl **953** is made of metal and is lower in height than the operator's seat **940**. The cowl **953** may serve to protect the operator's feet from exposure to any materials being lifted on the lift carriage **960** which may fall into the operator's compartment **905**. In an embodiment, the cowl **953** is higher than the height of the seat **940**.

[0174] FIG. **28** is an illustration of an embodiment of a lift truck **900** having seatside steering, a low operator compartment floor, a short front cowl **953** and multiple foot controls **955**. The embodiments of the lift truck **900** shown in FIG. **28** includes an overhead guard **970**, a lift chassis **960** having forks **965**, mast **910**, hand-operated controls **931** and **932** integrated into seat **940**, counterweight **912**, and a sloping rear panel **911**. The truck **900** has an operator's seat **940** that is not attached to the operator compartment floor **950**. Multiple foot pedals **955** are shown.

[0175] FIGS. **29** and **30** are illustrations of an embodiment of lift trucks **900** having a steering wheel **935**, a low operator compartment floor **950**, a tall front cowl **953**, and a single foot pedal **955**. The embodiments of the lift trucks shown in FIGS. **29-30** include varying form factors and have the lift assembly removed for clarity. The lift trucks **900** shown in

[0176] FIGS. **29-30** include an overhead guard **970**, an operator compartment **905**, four wheels, chassis **930**, hand-operated finger controls **932** integrated into seat **940**, and counterweight **912**. The lift trucks **900** shown in FIGS. **29** and **30** illustrate two examples of operator-accessible storage compartments, but other, different operator-accessible storage compartments are possible in the space available under the seat **940** or elsewhere in the open space that a broad, flat floor makes available.

[0177] Referring to FIG. **29**, the embodiment of truck **900** shown has an operator's seat **940** that

does not adjust vertically. Below the seat **940** and above the floor **950**, there is an operator-accessible storage compartment **990** that is open in the front. The tall front cowl **953** is made with an upper horizontal metallic portion and a transparent lower portion, such that the operator can view the lift carriage when the carriage is close to the ground.

[0178] Referring to FIG. **30**, the embodiment of the truck **900** shown also has an operator's seat **940** that does not adjust vertically and a tall front cowl **953** that is made with an upper horizontal metallic portion and a transparent lower portion. Below the seat **940** and above the floor **950**, there is an operator-accessible storage compartment **992** that is open in the front, and there are also open compartments to the side of the seat, for additional storage.

[0179] Additional examples of the advantageous lift truck elements described herein can be found in U.S. Design patent application Ser. No. 29/685,948, filed on Mar. 1, 2019, which is herein incorporated by reference. The lift trucks shown in U.S. Design patent application Ser. No. 29/685,948 include an optional step between the ground and the operator compartment floor so that operators who are accustomed to using a step may utilize one even though not necessary.

[0180] One will appreciate that some aspects of the lift truck disclosed herein can also be utilized for end-rider trucks, such as the modularity of configurations, the low floor, the interchangeable components, and/or the batteries and electric motors.

CONCLUSION

[0181] The terms and descriptions used above are set forth by way of illustration and example only and are not meant as limitations. Those skilled in the art will recognize that many variations, enhancements and modifications of the concepts described herein are possible without departing from the underlying principles of the invention. For example, skilled persons will appreciate that the subject matter of any sentence or paragraph can be combined with subject matter of some or all of the other sentences or paragraphs, except where such combinations are mutually exclusive. The scope of the invention should therefore be determined only by the following claims, claims presented in a continuation patent application, and equivalents to the foregoing claims.

Claims

1. A sit-down counterbalanced forklift truck comprising: a chassis having a front side, a rear side, a left side, and a right side; a chassis compartment situated between the front side and the left side of the chassis and between the left side and right side of the chassis; a power plant located in the chassis compartment; a plurality of wheels, at least one of which is operably connected to the power plant to drive the truck, wherein the plurality of wheels are sized and placed to provide the truck with a footprint and maneuverability suitable for use in a warehouse environment; a lift assembly attached to the chassis and having forks configured to lift and to lower a load, wherein the lift assembly is operably connected to the power plant; a counterweight attached to the rear side of the chassis; an operator compartment above the chassis and attached to the chassis; an operator seat in the operator compartment; and an operator compartment floor above the chassis compartment and along a bottom interior surface of the operator compartment, wherein the truck has a center of gravity situated at a gravitational center height that is lower than or equal to a height of at least one of the plurality of wheels.
2. A sit-down counterbalanced forklift truck according to claim 1, wherein the power plant comprises a motor operably connected to a battery located in the chassis compartment.
3. A sit-down counterbalanced forklift truck according to claim 2, further comprising a drive axle attached to one or more of the drive wheels, wherein the motor is located in the drive axle.
4. A sit-down counterbalanced forklift truck according to claim 2, further comprising: a hydraulic system in the chassis compartment, wherein the hydraulic system is operably connected to the battery.
5. A sit-down counterbalanced forklift truck according to claim 4, wherein the hydraulic system

comprises a hydraulic pump, a motor configured to operate the hydraulic pump, a hydraulic tank fluidly connected to the hydraulic pump, and a hydraulic valve, wherein the hydraulic pump is directly connected to the hydraulic tank without a hose.

6. A sit-down counterbalanced forklift truck according to claim 2, wherein the battery is a service-free battery.
7. A sit-down counterbalanced forklift truck according to claim 6, wherein the service-free battery is a lithium-based battery.
8. A sit-down counterbalanced forklift truck according to claim 1, wherein the chassis compartment is an operator-inaccessible compartment.
9. A sit-down counterbalanced forklift truck according to claim 8, wherein the operator compartment floor is not readily removable by the operator and comprises a cover for the operator-inaccessible chassis compartment.
10. A sit-down counterbalanced forklift truck according to claim 1, wherein the operator compartment floor is between 17 inches and 22 inches above the ground.
11. A sit-down counterbalanced forklift truck according to claim 1, wherein one or more of the plurality of wheels has a wheel height, and wherein the operator compartment floor has a floor height that is less than or equal to the wheel height.
12. A sit-down counterbalanced forklift truck according to claim 1, wherein one or more of the plurality of wheels has a wheel height, and wherein the operator compartment floor has a floor height that is less than or equal to 90% of the wheel height.
13. A sit-down counterbalanced forklift truck according to claim 1, wherein one or more of the plurality of wheels has a wheel height and a wheel radius, and the operator compartment floor has a floor height that is between the wheel radius and the wheel height.
14. A sit-down counterbalanced forklift truck according to claim 1, further comprising: an operator-accessible compartment having an operator-openable cover; and an operator-serviceable component located in the operator-accessible compartment.
15. A sit-down counterbalanced forklift truck according to claim 14, wherein the operator-serviceable component is one of a motor controller, fuse, VSM, contactor, or any combination thereof.
16. A sit-down counterbalanced forklift truck according to claim 1, wherein the operator compartment floor is at a height above ground such that an operator of a height within 95% of a human population can get in the operator compartment in one step from the ground and such that the operator can get out of the operator compartment with one step to the ground.
17. A sit-down counterbalanced forklift truck according to claim 1, wherein the chassis compartment has a chassis compartment top area between the front side and the rear side of the chassis and between the left side and right side of the chassis, and wherein the operator compartment floor has a floor area that is greater than or equal to 90% of the chassis compartment top area.
18. A sit-down counterbalanced forklift truck according to claim 1, wherein the chassis compartment has a chassis compartment top area between the front side and the rear side of the chassis and between the left side and right side of the chassis, and wherein the operator compartment floor has a floor area that is greater than or equal to 75% of the chassis compartment top area.
19. A sit-down counterbalanced forklift truck according to claim 1, wherein the operator compartment floor has a floor area that is substantially flat for greater than or equal to 50% of an area of a top of the chassis compartment.
20. A sit-down counterbalanced forklift truck according to claim 1, wherein the counterweight comprises multiple distinct pieces.
21. A sit-down counterbalanced forklift truck according to claim 1, wherein the operator compartment floor extends substantially flat side-to-side across approximately the entire width of

the truck.

22. A sit-down counterbalanced forklift truck according to claim 1, wherein the operator compartment floor is between 10 inches and 30 inches above the ground.

23. A sit-down counterbalanced forklift truck according to claim 1, wherein the operator seat is attached to the operator compartment floor.

24. A sit-down counterbalanced forklift truck according to claim 1, wherein the operator seat is rotatable about a vertical axis.

25. A sit-down counterbalanced forklift truck according to claim 24, wherein the operator seat has a range of rotation of greater than or equal to ± 180 degrees from a straight-forward-facing direction.

26. A sit-down counterbalanced forklift truck according to claim 24, wherein the operator seat has a range of rotation of greater than or equal to ± 22 degrees from a straight-forward-facing direction.

27. A sit-down counterbalanced forklift truck according to claim 24, wherein the operator seat is configured to slide backward as the seat rotates, so as to keep the operator's legs within an operator compartment as the seat rotates.

28. A sit-down counterbalanced forklift truck according to claim 1, wherein the operator seat has a bottom having front, left, right, and rear edges, and wherein the operator compartment floor extends substantially flat front-to-back from a front cowl proximate the front of the truck to a rear floor edge behind the front edge of the seat bottom.

29. A sit-down counterbalanced forklift truck according to claim 1, wherein there is an amount of leg room on the operator compartment floor to permit an operator to swing the operator's legs laterally while keeping the operator's feet above the floor.

30. A sit-down counterbalanced forklift truck according to claim 1, wherein the truck has no steering wheel extending from the operator compartment floor.

31. A sit-down counterbalanced forklift truck according to claim 1, wherein the truck has no foot controls.

32. A sit-down counterbalanced forklift truck according to claim 1, wherein the seat comprises at least one armrest, and wherein operator controls are located on said at least one armrest.

33. A sit-down counterbalanced forklift truck according to claim 1, further comprising: a steering wheel.

34. A sit-down counterbalanced forklift truck according to claim 1, further comprising: one or more foot controls.

35. A sit-down counterbalanced forklift truck according to claim 1, wherein operator compartment floor provides structural strength to the chassis to resist deflection of the chassis.

36. A sit-down counterbalanced forklift truck according to claim 1, wherein the plurality of wheels comprises at least one left wheel and at least one right wheel, and wherein the operator compartment floor extends side-to-side from the at least one left wheel to the at least one right wheel.

37. A sit-down counterbalanced forklift truck according to claim 1, wherein the plurality of wheels consists of three wheels.

38. A sit-down counterbalanced forklift truck according to claim 1, wherein the plurality of wheels consists of four wheels.

39. A sit-down counterbalanced forklift truck according to claim 1, wherein the truck employs multiple motors.

40. A sit-down counterbalanced forklift truck according to claim 1, wherein the truck has a center of gravity situated at a gravitational center height that is lower than or equal to a floor height of the operator compartment.

41. A sit-down counterbalanced forklift truck according to claim 1, wherein the operator compartment is equally accessible to an operator from both the right and left sides of the chassis.

42. A sit-down counterbalanced forklift truck according to claim 1, wherein the operator compartment has an operator compartment height from the floor to an overhead guard that is

greater than or equal to 5 feet.

43. A sit-down counterbalanced forklift truck according to claim 1, wherein the operator compartment has an operator compartment area between the front side and the rear side of the chassis and between the left side and right side of the chassis, and the operator compartment floor has a floor area that is greater than or equal to 85% of the operator compartment area.

44. A sit-down counterbalanced forklift truck according to claim 1, further comprising: an overhead guard having an overhead-guard area, and the operator compartment floor has a floor area that is greater than or equal to 85% of the overhead-guard area.

45. A sit-down counterbalanced forklift truck according to claim 1, wherein the truck is for use in an indoor warehouse on a warehouse floor, and wherein the plurality of wheels are sized and placed to provide the truck with a footprint and maneuverability suitable for use in a warehouse environment.

46. A sit-down counterbalanced forklift truck according to claim 45, wherein there is no intermediate step on the truck between the warehouse floor and the operator compartment floor to facilitate entry into or exit from the operator compartment by an operator.

47. A sit-down counterbalanced forklift truck according to claim 1, wherein the operator seat comprises a backrest, and wherein the entire backrest of the operator seat is above the operator compartment floor.
